$\boxed{\textbf{HW 7 - Zeshui Song}}$

Element Impedances in the Laplace Domain: $V = IZ$

- Resistor: $Z = R$
- Inductor: $Z = Ls$
- Capacitor: $Z = \frac{1}{Cs}$

<u>Note:</u> We also have impedances in the frequency domain. Substituting $s = j\omega$ (integration and differentiation in the time domain corresponds to multiplication by $j\omega$ in the frequency domain), we have:

- Resistor: $Z = R$
- Inductor: $Z = j\omega L$
- Capacitor: $Z = \frac{1}{j\omega C}$

By finding the equivalent impedance of a circuit, we can find the transfer function $H(s) = V(s)/I(s) = Z_{\text{eq}}$. Where the denominator of the transfer function gives us the characteristic equation, which we can use to find the natural frequency and damping ratio of the system.

Impedances add like resistors:

- Series: $Z_{\text{eq}} = Z_1 + Z_2 + \cdots + Z_n$
- Parallel: $\frac{1}{Z_{\text{eq}}} = \frac{1}{Z_1} + \frac{1}{Z_2} + \cdots + \frac{1}{Z_n}$

Admittances in the Laplace Domain: $I = YV$ Admittance is the reciprocal of impedance

$$Y = \frac{1}{Z}$$

They add like conductances:

- Series: $\frac{1}{Y_{\text{eq}}} = \frac{1}{Y_1} + \frac{1}{Y_2} + \cdots + \frac{1}{Y_n}$
- Parallel: $Y_{\text{eq}} = Y_1 + Y_2 + \cdots + Y_n$

Element Relations in the Time Domain:

- **Resistor**

$$v_R = iR$$
$$i(t) = \frac{1}{R}(v_1 - v_2)$$

- **Capacitor**

$$q = Cv_c \implies v_c = \frac{1}{C}\int i(t)dt$$
$$\frac{d}{dt}(q) = \frac{d}{dt}(Cv_c) \implies i(t) = C\frac{dv_C}{dt}$$

- **Inductor**

$$v_L = L\frac{di}{dt}$$
$$i(t) = i_o + \frac{1}{L}\int_0^t v_L(t')dt'$$

Defining Frequencies and Damping For a RLC circuit:

- Natural Frequency: $\omega_0 = \frac{1}{\sqrt{LC}}$
- Exponential damping coefficient: $\alpha = \frac{1}{2RC}$ (parallel RLC), $\alpha = \frac{R}{2L}$ (series RLC)
- Damping Ratio: $\zeta = \alpha/\omega_0$

Note: Overdamped: $\zeta > 1$, Critically damped: $\zeta = 1$, Underdamped: $\zeta < 1$.

We can also compare the characteristic equation of the RLC circuit to the standard second order system:

$$s^2 + 2\zeta\omega_0 s + \omega_0^2 = 0 \quad \text{or} \quad s^2 + 2\alpha s + \omega_0^2 = 0$$

Solving for the Time Domain Response:

Response Type	**Condition**	**General Solution (for both $i(t)$ and $v(t)$)**
Overdamped	$\alpha > \omega_0$	$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$
Critically Damped	$\alpha = \omega_0$	$i(t) = e^{-\alpha t}(A_1 t + A_2)$
Underdamped	$\alpha < \omega_0$	$i(t) = e^{-\alpha t}(B_1 \cos\omega_d t + B_2 \sin\omega_d t), \quad \omega_d = \sqrt{\omega_0^2 - \alpha^2}$

Note: The exponents s_1 and s_2 are simply $-\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$. Solve for the constants A_1, A_2, B_1, and B_2 using the initial conditions of the problem.

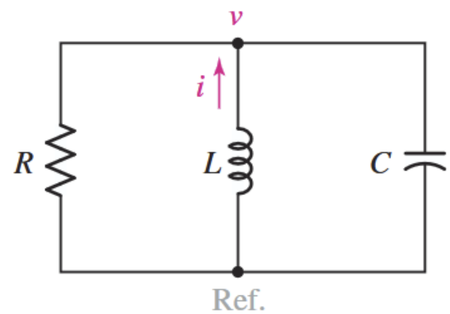
In the case of a DRIVEN RLC circuit, the total response is the sum of the natural response and the forced response (steady state value for i or v).

9.2

2. Element values of 10 mF and 2 nH are employed in the construction of a simple source-free parallel *RLC* circuit. (*a*) Select *R* so that the circuit is just barely overdamped. (*b*) Write the equation for the resistor current if its initial value is $i_R(0^+) = 13$ pA and $d\,i_R/dt|_{t=0^+} = 1$ nA/s.

Given:

- $C = 10$ mF
- $L = 2$ nH



(a) Find R such that the circuit is just barely overdamped.

For an overdamped circuit, $\alpha > \omega_0$. Substituting the values of α and ω_0, we have:

$$\frac{1}{2RC} > \frac{1}{\sqrt{LC}}$$

$$\implies R < \frac{1}{2}\sqrt{\frac{L}{C}} = \frac{1}{2}\sqrt{\frac{[2 \times 10^{-9}\,\text{H}]}{[10 \times 10^{-3}\,\text{F}]}} = 223.606 \times 10^{-6}\,\Omega$$

Thus, just barely less than $223.606\,\mu\Omega$,

$$\boxed{R = 223.5 \times 10^{-6}\,\Omega}$$

(b) Find $i_R(t)$ given $i_R(0^+) = 13$ pA, and $di_R/dt(0^+) = 1$ nA/s.

Since the circuit is overdamped, the response is of the form:

$$i_R(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Where s_1 and s_2 are the roots of the characteristic equation:

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$\alpha = \frac{1}{2RC} = \frac{1}{2[223.5 \times 10^{-6}\,\Omega][10 \times 10^{-3}\,\text{F}]} = 223.71 \times 10^3$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{[2 \times 10^{-9}\,\text{H}][10 \times 10^{-3}\,\text{F}]}} = 223.6 \times 10^3$$

$$s_1 = -230.72 \times 10^3, \quad s_2 = -216.69 \times 10^3$$

Now solving for A_1 and A_2 using the initial conditions:

- $i_R(0^+) = 13 \times 10^{-12}$ A
$$i_R(0^+) = A_1 + A_2 = 13 \times 10^{-12}$$
- $\frac{di_R}{dt}\Big|_{t=0^+} = 1 \times 10^{-9}$ A/s
$$\frac{di_R}{dt}(0^+) = A_1 s_1 + A_2 s_2 = 1 \times 10^{-9}$$

Solving the system of equations, we have:

$$A_1 = 2.13 \times 10^{-10}, \quad A_2 = -2.00 \times 10^{-10}$$

Thus, the time domain response is:

$$\boxed{i_R(t) = (2.13 \times 10^{-10})e^{-230.72\times 10^3 t} + (-2.00 \times 10^{-10})e^{-216.69\times 10^3 t} \text{ A}}$$

Note: We could also solve this using laplace transforms using the governing equation (KCL):

$$\frac{v}{R} + \left(i_0 + \frac{1}{L}\int_0^t v(t')\,dt'\right) + C\frac{dv}{dt} = 0$$

$$\implies V(s) = \frac{RLCv_0 s - RLi_0}{RLCs^2 + Ls + R}$$

Finding the roots of the denominator gives us s_1 and s_2:

$$V(s) = v_0\left[\frac{s}{(s-s_1)(s-s_2)}\right] - \frac{i_0}{C}\left[\frac{1}{(s-s_1)(s-s_2)}\right]$$

Compare to form:

$$\frac{s}{(s+a)(s+b)} \to \frac{1}{b-a}(be^{-bt} - ae^{-at})$$
$$\frac{1}{(s+a)(s+b)} \to \frac{1}{b-a}(e^{-at} - e^{-bt})$$

Be careful with the lapace transform of the integral term. Since we already included the initial condition i_0, we can assume the integral has zero initial conditions when taking the laplace transform, so $\mathcal{L}\left\{\int_0^t v(t')\,dt'\right\} = \frac{V(s)}{s}$, not $\frac{V(s)}{s}$ + initial conditions.

9.13

13. Consider the circuit depicted in Fig. 9.40. (*a*) Obtain an expression for $i_L(t)$ valid for all $t > 0$. (*b*) Obtain an expression for $i_R(t)$ valid for all $t > 0$. (*c*) Determine the settling time for both i_L and i_R.

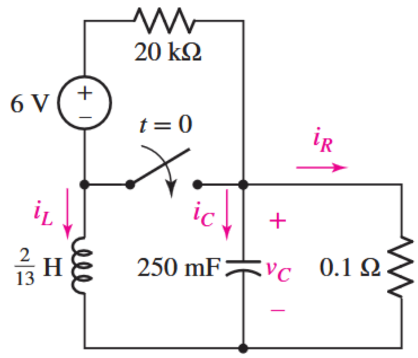
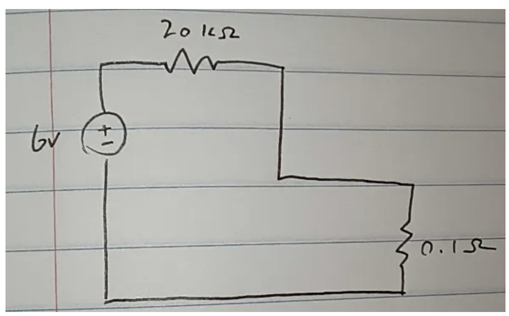


FIGURE 9.40

Initial conditions ($t < 0$) at steady state: the capacitor is an open circuit and the inductor is a short circuit.



Using voltage division, we can find the voltage across the 0.1 Ω resistor (which is equal to v_C):

$$v_{0.1\Omega} = \frac{[0.1\,\Omega]}{[0.1 + 20 \times 10^3\,\Omega]}[6\,\mathrm{V}] = 29.99 \times 10^{-6}, \mathrm{V}$$

The current through the resistors is equal to i_L:

$$i = \frac{[6\,\mathrm{V}]}{[0.1 + 20 \times 10^3\,\Omega]} = 299.99 \times 10^{-6}\,\mathrm{A}$$

Thus the initial conditions are:

- $i_R(0^-) = i_L(0^-) = 299.99 \times 10^{-6}$ A, $i_C(0^-) = 0$ A
- $v_{0.1\Omega}(0^-) = v_C(0^-) = 29.99 \times 10^{-6}$ V, $v_L(0^-) = 0$ V

After the switch is closed, current from the 6V source will follow the path of least resistance, thus it will flow through the switch back to the source, and not through the rest of the circuit. We can therefore redraw the circuit for $t > 0$ as a source free parallel RLC circuit.

(a) Find $i_L(t)$ for $t > 0$.

First, we determine if the circuit is overdamped, underdamped, or critically damped. We have:

$$\alpha = \frac{1}{2RC} = \frac{1}{2[0.1\,\Omega][250 \times 10^{-3}\,\text{F}]} = 20$$
$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{[0.1538\,\text{H}][250 \times 10^{-3}\,\text{F}]}} = 5.099$$

Since $\alpha > \omega_0$, the circuit is overdamped, and the response is of the form:

$$i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Where s_1 and s_2 are the roots of the characteristic equation:

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$
$$s_1 = -39.33, \quad s_2 = -0.66$$

Now solving for A_1 and A_2 using the initial conditions:

- $i_L(0^+) = 299.99 \times 10^{-6}$ A

$$i_L(0^+) = A_1 + A_2 = 299.99 \times 10^{-6}$$

- $v_L(0^+) = L\,\frac{di_L}{dt}\big|_{t=0^+} = v_C(0^+) \implies \frac{di_L}{dt}\big|_{t=0^+} = 194.93 \times 10^{-6}$ A/s

$$\left.\frac{di_L}{dt}\right|_{t=0^+} = A_1 s_1 + A_2 s_2 = 194.93 \times 10^{-6}$$

Note: Although voltage across the capacitor can't change instantaneously, the current through the inductor can change instantaneously, thus when the switch closes, v_L will jump from 0V to $v_C(0^+)$.

Solving the system of equations, we have:

$$A_1 = -1.02 \times 10^{-5}, \quad A_2 = 3.10 \times 10^{-4}$$

Thus, the time domain response is:

$$\boxed{i_L(t) = (-1.02 \times 10^{-5})e^{-39.33t} + (3.10 \times 10^{-4})e^{-0.66t}\,\text{A}}$$

(b) Find $i_R(t)$ for $t > 0$.

We can first find $v_R(t) = v_C(t)$ since they are in parallel, and then use Ohm's law to find $i_R(t)$. Since we know that the circuit is overdamped, the form of the response is the same as $i_L(t)$, so we can write:

$$V_C(t) = B_1 e^{s_1 t} + B_2 e^{s_2 t}$$

Where s_1 and s_2 are the same as before. Now solving for B_1 and B_2 using the initial conditions:

- $v_C(0^+) = 29.99 \times 10^{-6}$ V

$$v_C(0^+) = B_1 + B_2 = 29.99 \times 10^{-6}$$

- $\frac{dv_C}{dt}\Big|_{t=0^+}$:

  Using KCL at the top node, we have:

  $$i_R(0^+)+i_L(0^+)+i_C(0^+) = 0 \implies i_C(0^+) = -\frac{v}{R} - [299.99 \times 10^{-6}\text{ A}] = -599.98 \times 10^{-6}\text{ A}$$

  $$i_C(0^+) = C\,\frac{dv_C}{dt}\bigg|_{t=0^+} \implies \frac{dv_C}{dt}\bigg|_{t=0^+} = -23.99 \times 10^{-4}\text{ V/s}$$

Solving the system of equations, we have:

$$B_1 = 6.15 \times 10^{-5}, \quad B_2 = -3.15 \times 10^{-5}$$

Thus, the time domain response for $i_R(t) = \frac{1}{R}V_C(t)$ is:

$$\boxed{i_R(t) = \frac{1}{0.1}\left(6.15 \times 10^{-5}e^{-39.33t} - 3.15 \times 10^{-5}e^{-0.66t}\right)\text{ A}}$$

(c) Find the settling time for $i_L(t)$ and $i_R(t)$.

Settling time is the time it takes for the response to decay and remain below 1% of its maximum value $|i_m|$.

For an overdamped response, we can neglect the faster root and look at the slower root (the one closer to zero):

$$i_L(t) \approx 3.1 \times 10^{-4}e^{-0.66t} \implies i_L(t_s) = 0.01 \cdot 3.1 \times 10^{-4} = 3.1 \times 10^{-4}e^{-0.66t_s}$$

$$\implies \boxed{t_s = 6.97\,\text{s} \quad (\text{for } i_L(t))}$$

Similarly, for $i_R(t)$, we have:

$$i_R(t) \approx \frac{1}{0.1}\left(-3.15 \times 10^{-5}e^{-0.66t}\right) \implies i_R(t_s) = 0.01 \cdot \frac{1}{0.1}\left|-3.15 \times 10^{-5}\right| = \frac{1}{0.1}\left|-3.15 \times 10^{-5}e^{-0.66t_s}\right|$$

$$\implies \boxed{t_s = 6.97\,\text{s} \quad (\text{for } i_R(t))}$$

Since they share the same root, they have the same settling time.

9.16 16. With regard to the circuit presented in Fig. 9.42, (*a*) obtain an expression for $v(t)$ which is valid for all $t > 0$; (*b*) calculate the maximum inductor current and identify the time at which it occurs; (*c*) determine the settling time.

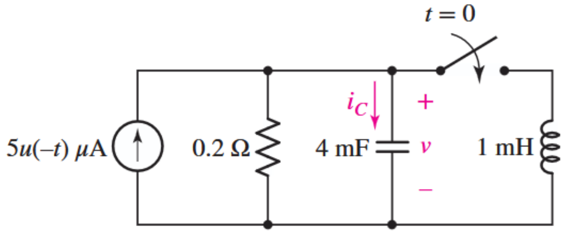


FIGURE 9.42

Notice that the current source is $5u(-t)$ which means that the current source is only on for $t < 0$, and turns off at $t = 0$!

For $t < 0$, at steady state (capacitor is an open circuit, inductor is disconnected):

- $i_L(0^-) = 0$ A, $v_L(0^-) = 0$ V
- $i_R(0^-) = 5\,\mu$A, $v_R(0^-) = [5\,\mu\text{A}][0.2\,\Omega] = 1 \times 10^{-6}$ V
- $i_C(0^-) = 0$ A, $v_C(0^-) = 1 \times 10^{-6}$ V

For $t > 0$, the current source turns off, and the inductor is now connected. Since the capacitor voltage doesn't change instantaneously:

- $v_R(0^+) = v_C(0^+) = v_L(0^+) = 1 \times 10^{-6}$ V
- $i_R(0^+) = 5\,\mu$A
- Since the inductor current cannot change instantaneously, $i_L(0^+) = i_L(0^-) = 0$ A
- $\frac{dv}{dt}\Big|_{t=0^+} = \frac{1}{C} i_C(0^+)$: Using KCL at the top node, we have:

$$i_R(0^+) + i_L(0^+) + i_C(0^+) = 0 \implies i_C(0^+) = -i_R(0^+) = -5\,\mu\text{A}$$

$$\implies \frac{dv}{dt}\bigg|_{t=0^+} = \frac{[-5 \times 10^{-6}\,\text{A}]}{[4 \times 10^{-3}\,\text{F}]} = -0.00125\,\text{V/s}$$

(a) Find $v(t)$ for $t > 0$.

First, we determine if the circuit is overdamped, underdamped, or critically damped. We have:

$$\alpha = \frac{1}{2RC} = \frac{1}{2[0.2\,\Omega][4 \times 10^{-3}\,\text{F}]} = 625$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{[10^{-3}\,\text{H}][4 \times 10^{-3}\,\text{F}]}} = 500$$

Since $\alpha > \omega_0$, the circuit is overdamped, and the response is of the form:

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

where s_1 and s_2 are the roots of the characteristic equation.

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$
$$s_1 = -1000, \quad s_2 = -250$$

Now solving for A_1 and A_2 using the initial conditions (using v_C since everything is in parallel):

- $v_C(0^+) = 1 \times 10^{-6}$ V
$$v_C(0^+) = A_1 + A_2 = 1 \times 10^{-6}$$
- $\frac{dv_C}{dt}\Big|_{t=0^+} = -0.00125$ V/s
$$\frac{dv_C}{dt}\bigg|_{t=0^+} = A_1 s_1 + A_2 s_2 = -0.00125$$

Solving the system of equations, we have:

$$A_1 = 1.33 \times 10^{-6}, \quad A_2 = -3.33 \times 10^{-7}$$

Thus, the time domain response is:

$$\boxed{v(t) = (1.33 \times 10^{-6})e^{-1000t} + (-3.33 \times 10^{-7})e^{-250t}\,\text{V}}$$

(b) Find the max inductor current and the time at which it occurs.

The inductor current is given by

$$i_L(t) = i_L(0^+) + \int_0^t \frac{v_L(t')}{L} dt'$$

Since $i_L(0^+) = 0$ A and $v_L(t) = v(t)$, we have:

$$i_L(t) = \frac{1}{10^{-3}} \int_0^t ((1.33 \times 10^{-6})e^{-1000t'} + (-3.33 \times 10^{-7})e^{-250t'})\, dt'$$
$$i_L(t) = 1.33 \times 10^{-6}(e^{-250t} - e^{-1000t})$$

The current is maximum when $\frac{di_L}{dt} = 0$:

$$\frac{di_L}{dt} = 1.33 \times 10^{-6}(-250e^{-250t} + 1000e^{-1000t}) = 0 \implies \boxed{t = 1.848 \times 10^{-3}\,\text{s}}$$

Substituting back into $i_L(t)$, we have:

$$\boxed{i_L(1.848 \times 10^{-3}\,\text{s}) = 0.628 \times 10^{-6}\,\text{A}}$$

(c) Settling time for $v(t)$.

$$v(t) \approx -3.33 \times 10^{-7} e^{-250t} \implies v(t_s) = 0.01 \cdot |-3.33 \times 10^{-7}| = |-3.33 \times 10^{-7} e^{-250t_s}|$$
$$\implies \boxed{t_s = 0.0184\,\text{s}}$$

9.21 21. A motor coil having an inductance of 8 H is in parallel with a 2 μF capacitor and a resistor of unknown value. The response of the parallel combination is determined to be critically damped. (*a*) Determine the value of the resistor. (*b*) Compute α. (*c*) Write the equation for the current flowing into the resistor if the top node is labeled v, the bottom node is grounded, and $v = Ri_r$. (*d*) Verify that your equation is a solution to the circuit differential equation,

$$\frac{di_r}{dt} + 2\alpha\frac{di_r}{dt} + \alpha^2 i_r = 0$$

(a) For a critically damped parallel RLC circuit, we have $\alpha = \omega_0$.

$$\implies \frac{1}{2RC} = \frac{1}{\sqrt{LC}}$$

$$R = \frac{1}{2}\sqrt{\frac{L}{C}} = \frac{1}{2}\sqrt{\frac{[8\,\mathrm{H}]}{[2\times 10^{-6}\,\mathrm{F}]}} = \boxed{1000\,\Omega}$$

(b) Find α

$$\alpha = \frac{1}{2RC} = \frac{1}{2[1000\,\Omega][2\times 10^{-6}\,\mathrm{F}]} = \boxed{250\,\mathrm{rad/s}}$$

(c) Find i_R:

Since the circuit is critically damped, the response is of the form:

$$v(t) = e^{-\alpha t}(A_1 t + A_2)$$

Where α is the same as before.

(d) Show that it is a solution to the differential equation:

$$\frac{d^2 i_R}{dt^2} + 2\alpha\frac{di_R}{dt} + \alpha^2 i_R = 0$$

Find derivatives:

$$\begin{aligned}\frac{di_R}{dt} &= A_1 e^{-\alpha t} - \alpha(A_1 t + A_2)e^{-\alpha t} \\ &= e^{-\alpha t}\left[A_1 - \alpha(A_1 t + A_2)\right]\end{aligned}$$

$$\begin{aligned}\frac{d^2 i_R}{dt^2} &= -\alpha e^{-\alpha t}\left[A_1 - \alpha(A_1 t + A_2)\right] + e^{-\alpha t}[-\alpha A_1] \\ &= e^{-\alpha t}\left[-\alpha A_1 + \alpha^2(A_1 t + A_2) - \alpha A_1\right] \\ &= e^{-\alpha t}\left[-2\alpha A_1 + \alpha^2(A_1 t + A_2)\right]\end{aligned}$$

Plugging these results back into the original equation:

$$\begin{aligned}
\text{LHS} &= e^{-\alpha t}\left[-2\alpha A_1 + \alpha^2(A_1 t + A_2)\right] + 2\alpha e^{-\alpha t}\left[A_1 - \alpha(A_1 t + A_2)\right] + \alpha^2 e^{-\alpha t}(A_1 t + A_2) \\
&= e^{-\alpha t}\Big[-2\alpha A_1 + \alpha^2(A_1 t + A_2) + 2\alpha A_1 - 2\alpha^2(A_1 t + A_2) + \alpha^2(A_1 t + A_2)\Big] \\
&= e^{-\alpha t}\Big[(-2\alpha A_1 + 2\alpha A_1) + (\alpha^2 - 2\alpha^2 + \alpha^2)(A_1 t + A_2)\Big] \\
&= e^{-\alpha t}[0 + 0] = 0
\end{aligned}$$

9.26

26. For the circuit of Fig. 9.45, $i_s(t) = 30u(-t)$ mA. (*a*) Select R_1 so that $v(0^+) = 6$ V. (*b*) Compute $v(2\text{ ms})$. (*c*) Determine the settling time of the capacitor voltage. (*d*) Is the inductor current settling time the same as your answer to part (*c*)?

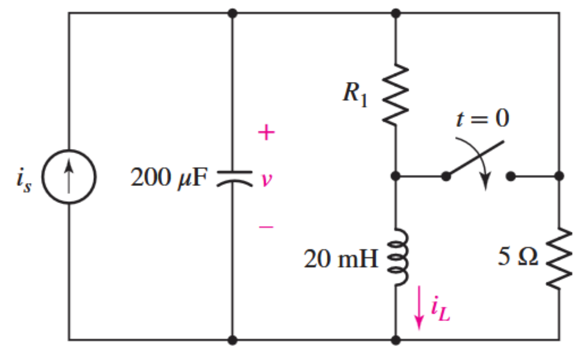
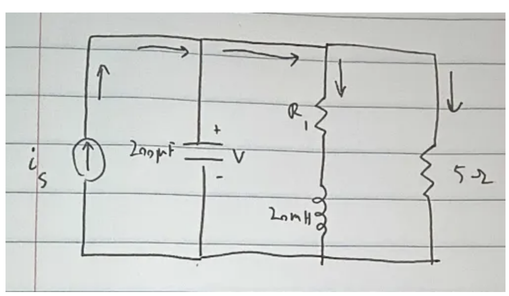


FIGURE 9.45

(a) Find R_1 such that $v(0^+) = 6$ V.

For $t < 0$ at steady state, the capacitor is an open circuit and the inductor is a short circuit and $i_s = 30$ mA.



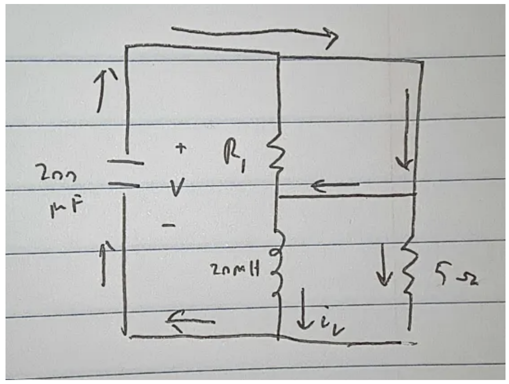
Thus,

$$R_{eq} = \frac{R_1 \cdot [5\,\Omega]}{R_1 + [5\,\Omega]}$$

$$v(0^-) = i_s \cdot R_{eq} = [30 \times 10^{-3}\,\text{A}] \cdot \frac{R_1 \cdot [5\,\Omega]}{R_1 + [5\,\Omega]} = 6\,\text{V}$$

$$\implies \frac{R_1 \cdot [5\,\Omega]}{R_1 + [5\,\Omega]} = 200$$

However, this is physically impossible. Looking at the circuit for $t > 0$:



Since the inductor maintains current in the same direction, the switch actually shorts R_1. Thus, it makes sense that we can't find a value of R_1 that satisfies the condition, since R_1 is effectively removed from the circuit for $t > 0$.

(b) Find $v(2\,\text{ms})$.

Without R_1, this is just a source free parallel RLC circuit. First we determine if the circuit is overdamped, underdamped, or critically damped. We have:

$$\alpha = \frac{1}{2RC} = \frac{1}{2[5\,\Omega][200 \times 10^{-6}\,\text{F}]} = 500$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{[20 \times 10^{-3}\,\text{H}][200 \times 10^{-6}\,\text{F}]}} = 500$$

Since $\alpha = \omega_0$, the circuit is critically damped, and the response is of the form:

$$v(t) = e^{-\alpha t}(A_1 t + A_2)$$

Now finding the initial conditions:

- Assuming that $v(0^+) = 6$ V
- $\frac{dv}{dt}\Big|_{t=0^+} = \frac{1}{C} i_C(0^+)$:

  Using KCL at the top node, we have:

  $$i_R(0^+) + i_L(0^+) + i_C(0^+) = 0 \implies i_C(0^+) = -\frac{v}{R} - i_L(0^+) = -\frac{[6\,\text{V}]}{[5\,\Omega]} - [30 \times 10^{-3}\,\text{A}] = -1.23\,\text{A}$$

  Since the inductor current cannot change instantaneously, $i_L(0^+) = i_L(0^-) = 30$ mA. Thus,

  $$\implies \frac{dv}{dt}\Big|_{t=0^+} = \frac{[-1.23\,\text{A}]}{[200 \times 10^{-6}\,\text{F}]} = -6150\,\text{V/s}$$

Solving for A_1 and A_2:

- $v(0^+) = 6$ V

$$v(0^+) = A_2 = 6\,\text{V}$$

- $\left.\frac{dv}{dt}\right|_{t=0^+} = -6150$ V/s

$$\left.\frac{dv}{dt}\right|_{t=0^+} = A_1 - \alpha A_2 = -6150$$

$$\implies A_1 = -3150, \quad A_2 = 6$$

Thus, the time domain response is:

$$\boxed{v(t) = e^{-500t}(-3150t + 6)\,\text{V}}$$

$$\implies \boxed{v(2\,\text{ms}) = -0.110\,\text{V}}$$

(c) Find the settling time of the capacitor voltage.

Since the circuit is critically damped, we can't just neglect a root. Thus we would need a numerical solver instead.

We find that

$$v_m = -0.894\,\text{V}$$

$$\boxed{t_s = 0.01718\,\text{s}}$$

(d) Is the inductor current settling time the same?

Yes, since the inductor current has the same form of response as the capacitor voltage, they will have the same settling time.

9.30

30. The circuit of Fig. 9.1 is constructed using component values 10 kΩ, 72 μH, and 18 pF. (*a*) Compute α, ω_d, and ω_0. Is the circuit overdamped, critically damped, or underdamped? (*b*) Write the form of the natural capacitor voltage response $v(t)$. (*c*) If the capacitor initially stores 1 nJ of energy, compute v at $t = 300$ ns.

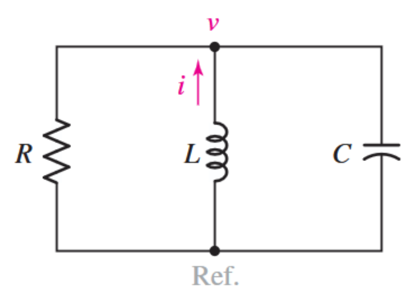


FIGURE 9.1 The source-free parallel *RLC* circuit.

(a) Find α, ω_0, and ω_d. First we determine if the circuit is overdamped, underdamped, or critically damped. We have:

$$\alpha = \frac{1}{2RC} = \frac{1}{2[10 \times 10^3\,\Omega][18 \times 10^{-12}\,\text{F}]} = \boxed{2.777 \times 10^6\,\text{rad/s}}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{[72 \times 10^{-6}\,\text{H}][18 \times 10^{-12}\,\text{F}]}} = \boxed{27.777 \times 10^6\,\text{rad/s}}$$

Since $\alpha < \omega_0$, the circuit is underdamped, and the damped natural frequency is given by:

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{(27.777 \times 10^6)^2 - (2.777 \times 10^6)^2} = \boxed{27.631 \times 10^6\,\text{rad/s}}$$

(b) The form of the response for $v(t)$ is:

$$\boxed{v(t) = e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)}$$

(c) Find $v(300\,\text{ns})$.

Given that the capacitor stores 1 nJ of energy at $t = 0$, we can find the initial voltage across the capacitor:

$$w = \frac{1}{2}Cv^2 \implies v(0^+) = \sqrt{\frac{2w}{C}} = \sqrt{\frac{2[1 \times 10^{-9}\,\text{J}]}{[18 \times 10^{-12}\,\text{F}]}} = 10.54\,\text{V}$$

Assuming there is zero current through the inductor at $t = 0^+$, we can find dv/dt at $t = 0^+$ using KCL at the top node:

$$i_R(0^+) + i_L(0^+) + i_C(0^+) = 0 \implies i_C(0^+) = -\frac{v(0^+)}{R}$$

$$i_C(0^+) = 0.00105\,\text{A} \implies \left.\frac{dv}{dt}\right|_{t=0^+} = \frac{i_C(0^+)}{C} = \frac{[0.00105\,\text{A}]}{[18 \times 10^{-12}\,\text{F}]} = 58.55 \times 10^6\,\text{V/s}$$

Now solving for B_1 and B_2:

- $v(0^+) = 10.54$ V

$$v(0^+) = B_1 = 10.54\,\text{V}$$

- $\left.\frac{dv}{dt}\right|_{t=0^+} = 58.55 \times 10^6$ V/s

$$\left.\frac{dv}{dt}\right|_{t=0^+} = -\alpha B_1 + \omega_d B_2 = 58.55 \times 10^6$$

Solving for B_1 and B_2, we have:

$$B_1 = 10.54, \quad B_2 = 3.175$$

Thus, the time domain response is:

$$\boxed{v(t) = e^{-2.777 \times 10^6 t}(10.54 \cos(27.631 \times 10^6 t) + 3.175 \sin(27.631 \times 10^6 t))\,\text{V}}$$

$$\implies \boxed{v(300\,\text{ns}) = -0.682, \text{V}}$$

9.35

35. For the circuit of Fig. 9.46, determine (*a*) $i_C(0^-)$; (*b*) $i_L(0^-)$; (*c*) $i_R(0^-)$; (*d*) $v_C(0^-)$; (*e*) $i_C(0^+)$; (*f*) $i_L(0^+)$; (*g*) $i_R(0^+)$; (*h*) $v_C(0^+)$.

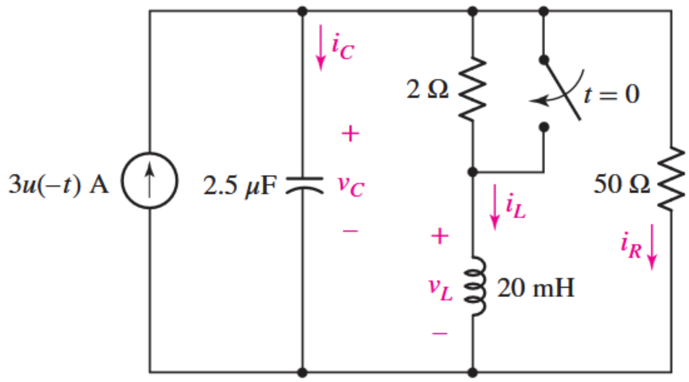
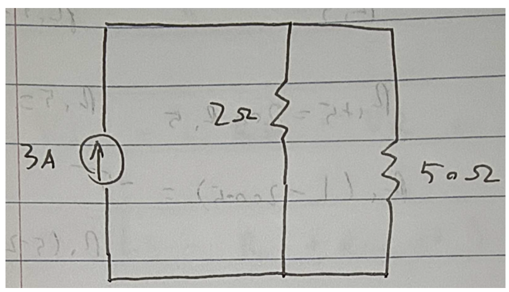


FIGURE 9.46

Circuit for $t < 0$:



Voltage across each parallel branch is then:

$$R_{eq} = \frac{[2\,\Omega]\cdot[50\,\Omega]}{[2+50\,\Omega]} = 1.92\,\Omega$$

$$V = [3\,\text{A}][1.92\,\Omega] = 5.76\,\text{V}$$

Current division for each branch:

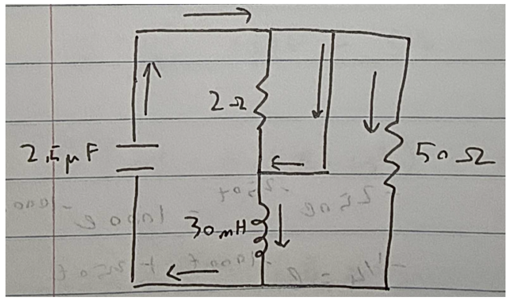
$$i_{2\Omega} = \frac{\frac{1}{[2\,\Omega]}}{\frac{1}{[1.92\,\Omega]}}[3\,\text{A}] = 2.88\,\text{A}$$

$$i_{50\Omega} = \frac{\frac{1}{[50\,\Omega]}}{\frac{1}{[1.92\,\Omega]}}[3\,\text{A}] = 0.115\,\text{A}$$

Therefore:

- $i_C(0^-) = 0$ A
- $i_L(0^-) = 2.88$ A
- $i_R(0^-) = 0.115$ A
- $v_C(0^-) = v_L(0^-) = v_R(0^-) = 5.76$ V

Circuit for $t > 0$:



Voltage across capacitors don't change instantaneously, and current through inductors don't change instantaneously. Thus:

$$i_R(0^+) = \frac{v(0^+)}{R} = \frac{[5.76\,\mathrm{V}]}{[50\,\Omega]} = 0.115\,\mathrm{A}$$

By KCL at the top node:

$$i_R(0^+)+i_L(0^+)+i_C(0^+) = 0 \implies i_C(0^+) = -i_R(0^+)-i_L(0^+) = -[0.115\,\mathrm{A}]-[2.88\,\mathrm{A}] = -2.995\,\mathrm{A}$$

Thus, the initial conditions for $t > 0$ are:

- $i_C(0^+) = -2.995$ A
- $i_L(0^+) = 2.88$ A
- $i_R(0^+) = 0.115$ A
- $v_C(0^+) = v_L(0^+) = v_R(0^+) = 5.76$ V

9.46

46. With reference to the circuit depicted in Fig. 9.49, calculate (*a*) α; (*b*) ω_0; (*c*) $i(0^+)$; (*d*)$di/dt|_{0^+}$; (*e*) $i(t)$ at $t = 6$ s.

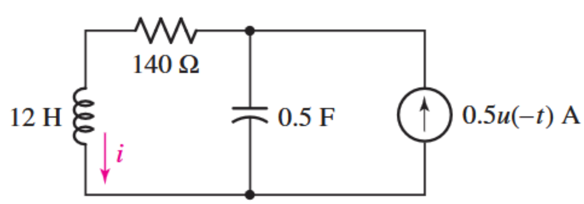


FIGURE 9.49

(a) Find α. (Series RLC circuit)

$$\alpha = \frac{R}{2L} = \frac{[140\,\Omega]}{2[12\,\mathrm{H}]} = \boxed{5.833\,\mathrm{rad/s}}$$

(b) Find ω_0.

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{[12\,\text{H}][0.5\,\text{F}]}} = \boxed{0.408\,\text{rad/s}}$$

Since $\alpha > \omega_0$, this circuit is overdamped.

(c) Find $i(0^+)$.

For $t < 0$ at steady state, the current through the inductor is equal to the source and since the inductor current cannot change instantaneously, we have:

$$\boxed{i(0^+) = i(0^-) = 0.5\,\text{A}}$$

(d) Find di/dt at $t = 0^+$.

Using the voltage over the inductor we can find di/dt at $t = 0^+$:

$$v_L(0^+) = L\left.\frac{di}{dt}\right|_{t=0^+}$$

At steady state at $t = 0^-$ the voltage drop across the resistor is equal to the voltage across the capacitor (they are in parallel), and since the voltage across the capacitor can't change instantaneously, we have:

$$v_C(0^-) = v_C(0^+) = v_R(0^-) = [140\,\Omega][0.5\,\text{A}] = 70\,\text{V}$$

Thus, subtracting the voltage drop across the resistor at $t = 0^+$ from the voltage across the capacitor, we can find the voltage across the inductor at $t = 0^+$:

$$v_L(0^+) = v_C(0^+) - v_R(0^+) = [70\,\text{V}] - [140\,\Omega][0.5\,\text{A}] = 0\,\text{A}$$

$$\implies \boxed{\left.\frac{di}{dt}\right|_{t=0^+} = 0\,\text{A/s}}$$

(e) Find $i(6)$.

Since the circuit is overdamped, the response is of the form:

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Where s_1 and s_2 are the roots of the characteristic equation:

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$s_1 = -11.65, \quad s_2 = -0.0142$$

Now solving for A_1 and A_2 using the initial conditions:

- $i(0^+) = 0.5$ A

$$i(0^+) = A_1 + A_2 = 0.5$$

- $\left.\frac{di}{dt}\right|_{t=0^+} = 0$ A/s

$$\left.\frac{di}{dt}\right|_{t=0^+} = A_1 s_1 + A_2 s_2 = 0$$

Solving the system of equations, we have:

$$A_1 = -6.1 \times 10^{-4}, \quad A_2 = 0.501$$

Thus, the time domain response is:

$$\boxed{i(t) = (-6.1 \times 10^{-4})e^{-11.65t} + (0.501)e^{-0.0142t}\ \text{A}}$$

$$\implies \boxed{i(6) = 0.46\ \text{A}}$$

9.50

50. The circuit in Fig. 9.52 has the switch in position *a* for a long time, with the capacitor discharged. At time $t = 0$, the switch is moved to position *b*. Determine the initial and final conditions for the capacitor and inductor (both current and voltage for each element).

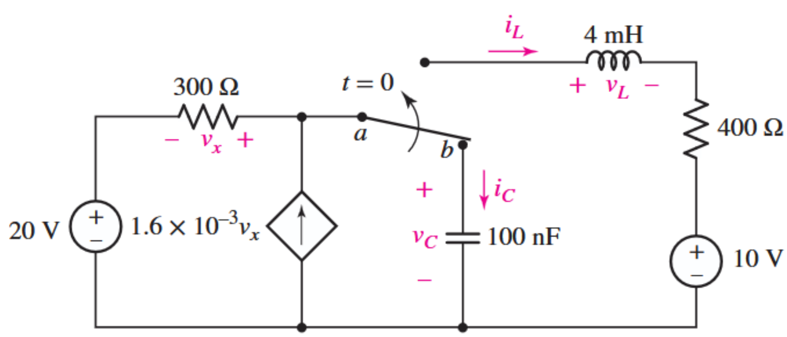
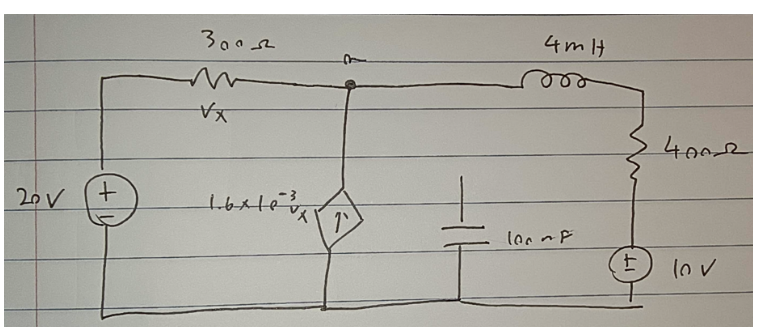


FIGURE 9.52

Note: This diagram is confusing so I'm going to be using my own interpretation of the circuit.

Circuit for $t < 0$:



Given that the capacitor is *discharged*, the voltage across the capacitor is 0V at $t = 0^-$. At steady state, the capacitor is also an open circuit, so there is no current.

KCL at the top node gives us:

$$v_x = v_a - 20$$

$$\frac{v_a - 20}{[300\,\Omega]} - 1.6 \times 10^{-3}(v_a - 20) + \frac{v_a - 10}{[400\,\Omega]} = 0$$

$$\implies v_a = 14.094\,\text{V}$$

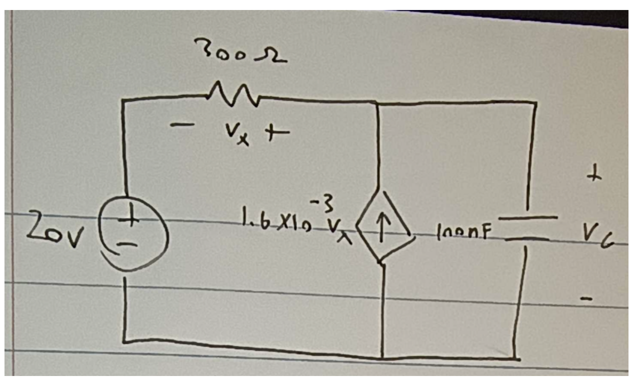
Thus, the current going through the 400 Ω resistor is:

$$i = \frac{v_a - 10}{[400\,\Omega]} = 0.0102\,\text{A}$$

Since the inductor is a short circuit in series with the 400 Ω resistor, the current through the inductor is also 0.0102 A. Thus, the initial conditions for $t < 0$ are:

- $i_L(0^-) = 0.0102$ A, $v_L(0^-) = 0$ V
- $i_C(0^-) = 0$ A, $v_C(0^-) = 0$ V

Circuit for $t > 0$:



At steady state, the inductor is completely cut off, so it has zero current and voltage. The capacitor is an open circuit with zero current, but we can find its voltage.

Using KCl at the top node:

$$\frac{v_a - 20}{[300\,\Omega]} - 1.6 \times 10^{-3}(v_a - 20) = 0$$

$$\implies v_a = 20\,\text{V}$$

Thus, the voltage across the capacitor is also $v_a = 20$ V.

- $i_L(0^+) = 0$ A, $v_L(0^+) = 0$ V
- $i_C(0^+) = 0$ A, $v_C(0^+) = 20$ V

9.54 54. Consider the circuit depicted in Fig. 9.55. If $v_s(t) = -8 + 2u(t)$ V, determine (*a*) $v_C(0^+)$; (*b*) $i_L(0^+)$; (*c*) $v_C(\infty)$; (*d*) $v_C(t = 150\text{ ms})$.

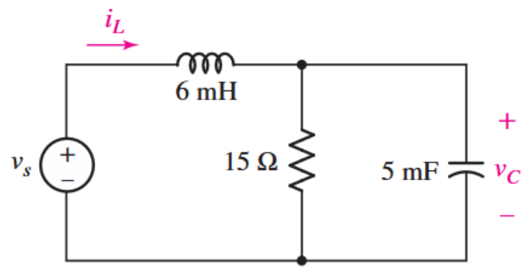


FIGURE 9.55

The voltage source has 2 states:

- $v_s = -8$ V for $t < 0$
- $v_s = -6$ V for $t > 0$

(a) Find $v_C(0^+)$.

For $t < 0$, at steady state:

$$v_s = v_C(0^-) = -8\,\text{V}$$

Since the voltage across the capacitor cannot change instantaneously, we have:

$$\boxed{v_C(0^+) = v_C(0^-) = -8\,\text{V}}$$

(b) Find $i_L(0^+)$.

Since the inductor current cannot change instantaneously, we can use $i_L(0^-)$. Since the inductor is in series with the 15 Ω resistor, we can find the current through the inductor at $t = 0^-$ using Ohm's law:

$$i_L(0^-) = \frac{v_s}{R} = \frac{[-8\,\text{V}]}{[15\,\Omega]} = -0.533\,\text{A}$$

$$\boxed{i_L(0^+) = i_L(0^-) = -0.533\,\text{A}}$$

(c) Find $v_C(\infty)$.

Same steps but with new value of v_s:

$$\boxed{v_s = v_C(\infty) = -6\,\text{V}}$$

(d) Find $v_C(150\,\text{ms})$.

Solving for the FREE response:
First, we determine if the circuit is overdamped, underdamped, or critically damped. Without the voltage source, it would be a parallel RLC circuit:

$$\alpha = \frac{1}{2RC} = \frac{1}{2[15\,\Omega][5\times10^{-3}\,\text{F}]} = 6.66\,\text{rad/s}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{[6\times10^{-3}\,\text{H}][5\times10^{-3}\,\text{F}]}} = 182.57\,\text{rad/s}$$

Since $\alpha < \omega_0$, the circuit is underdamped, and the response is of the form:

$$v_C(t) = e^{-\alpha t}(B_1\cos\omega_d t + B_2\sin\omega_d t) + v_C(\infty)$$

Where $\omega_d = \sqrt{\omega_0^2 - \alpha^2} = 182.45\,\text{rad/s}$.

Finding initial conditions:

$$v_C(0^+) = v_C(0^-) = -8\,\text{V}$$

Using KCL at the top node:

$$i_L(0^+) + i_R(0^+) + i_C(0^+) = 0 \implies i_C(0^+) = -i_L(0^+) - i_R(0^+)$$

$$i_C(0^+) = -[-0.533\,\text{A}] - [-0.533\,\text{A}] = 1.066\,\text{A}$$

$$\implies \frac{dv_C}{dt}(0^+) = \frac{i_C(0^+)}{C} = \frac{[1.066\,\text{A}]}{[5\times10^{-3}\,\text{F}]} = 213.2\,\text{V/s}$$

Now solving for B_1 and B_2:

- $v_C(0^+) = -8$ V

$$v_C(0^+) = B_1 = -8$$

- $\frac{dv_C}{dt}\Big|_{t=0^+} = 213.2$ V/s

$$\left.\frac{dv_C}{dt}\right|_{t=0^+} = -\alpha B_1 + \omega_d B_2 = 213.2$$

$$\implies B_1 = -8, \quad B_2 = 0.876$$

Thus, the FREE response is:

$$v_{C,0}(t) = e^{-6.66t}(-8\cos(182.45t) + 0.876\sin(182.45t))$$

Finding the TOTAL response
The total response is the sum of the free response and the forced response (steady state value $v_C(\infty)$):

$$\boxed{v_C(t) = e^{-6.66t}(-8\cos(182.45t) + 0.876\sin(182.45t)) - 6\,\text{V}}$$

$$\implies \boxed{v_C(150\,\text{ms}) = -3.93\,\text{V}}$$

Check this solution